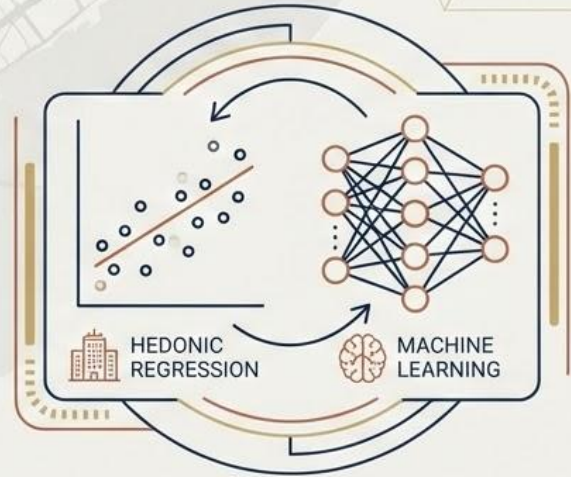


INTRODUCTION

# Data-Driven Property Valuation Factors and Model for Cebu City

Comparative Analysis of Hedonic Regression and Machine Learning Approaches



# Background

## Cebu Economic Growth & Context

- Cebu is the fastest-growing regional economy (7.3% growth in 2024).
- Real estate contributes ~6% to GDP, driven by IT-BPM and Tourism.



Cebu Economic Growth Rate



## The Problem: The valuation landscape is fragmented.

### Fragmented Valuation Landscape



**BIR Zonal Values:**  
Outdated (2018/2019) and static.



**Bank Appraisals:**  
Conservative and opaque.



**Market Listings:**  
Highly variable and often inflated.

# Statement of the Problem

## Statement of the Problem



There is **no scientifically validated, data-driven model** for residential property valuation in Cebu City.

Stakeholders (Developers, Brokers, Buyers) rely on:



"Gut feel" /  
Intuition



Disjointed  
data sources



Manual "Sales  
Comparison" of  
just ~3 properties.

## Core Question:

How can real estate firms utilize data-driven models to predict property values accurately and consistently?



# Research Objectives

## Research Objectives



1. **Develop** a rigorous data processing pipeline to aggregate sparse property data.
2. **Compare** the accuracy of **Hedonic Regression** (Traditional) vs. **Machine Learning** (Random Forest, XGBoost).
3. **Quantify** the “Valuation Gap” between administrative pricing (BIR) and market reality.
4. **Measure** the value of “Future Factors” (e.g., proximity to planned Cebu BRT stations).

# Solution Requirements

## Solution Requirements



**Data Aggregation:** Application to scrape and parse listings from web portals (Ceiling Price) and bank foreclosures (Floor Price).



**Geospatial Engine:** Logic to calculate “Isochrones” and distances (Haversine) to key hubs (Ayala, IT Park, BRT).



**Modeling Engine:** Python-based environment to train and tune ML models (RF, XGBoost).



**Dashboard:** A prototype interface (Streamlit) for plotting values and “What-If” scenarios.

# Significance

## Significance



### **For CPRE (Brokerage):**

- Provides a standardized, defensible pricing tool for clients.



### **For Investors/Buyers:**

- Increases transparency and trust in “Fair Market Value”.



### **For Policymakers:**

- Highlights the lag in tax mapping (Zonal Values) vs. market growth.



### **Academic Contribution:**

- First major study to apply ML valuation specifically to the *Cebu* context.



# Scope and Delimitations

## Scope and Delimitations



### Scope:

- **Location:** Cebu City, Mandaue, Talisay.
- **Type:** Residential (House & Lot, Condominium, Vacant Lot).
- **Timeframe:** Active listings Q4 2024 - Q1 2026.



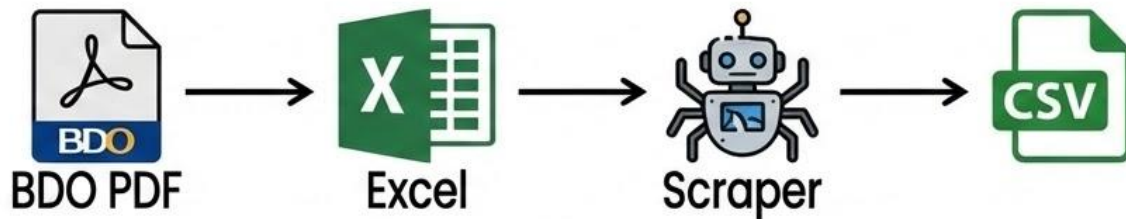
### Delimitations:

- Excludes Commercial/Industrial.
- No "Interior Finish" data.
- Uses Listing Prices as Market Value proxy.



# Dataset Source and Description

## Dataset Source and Description



### Hybrid Data Strategy:

1. **Floor Price (Verified)::** 900+ Foreclosed Listings from BDO (Conservative).
2. **Ceiling Price (Market)::** ~3,000+ Online Listings from Lamudi/DotProperty (Aggressive).

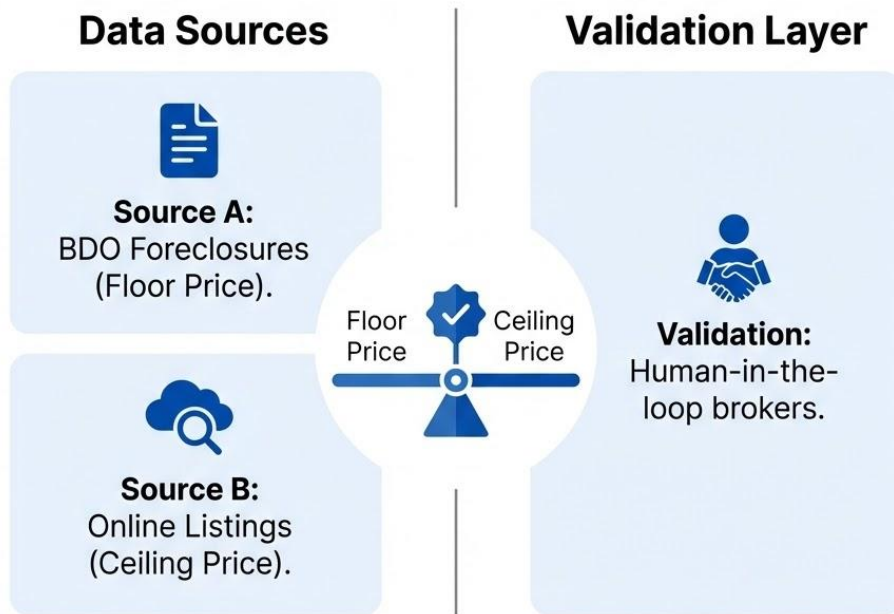
### Variables:

- **Target:** Price (Php) / Price per Sqm.
- **Features:** Lot Area, Floor Area, Bedrooms, Bathrooms, Location (Coords), Distance to CBD.



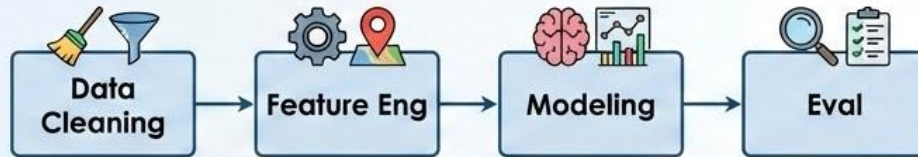
# Dataset Strategy

**Explanation:** Hybrid approach.



# Methodology

## Methodology



### 1. Preprocessing:

- Regex parsing ("3BR")
- Outlier Removal (IQR)
- Log-Transformation of Price.

### 2. Geospatial Feature Engineering:

- Calculate distance to nearest **Cebu BRT Station**.
- **Amenity Score**: Count of schools/hospitals within 1km radius.

### 3. Modeling Strategy:

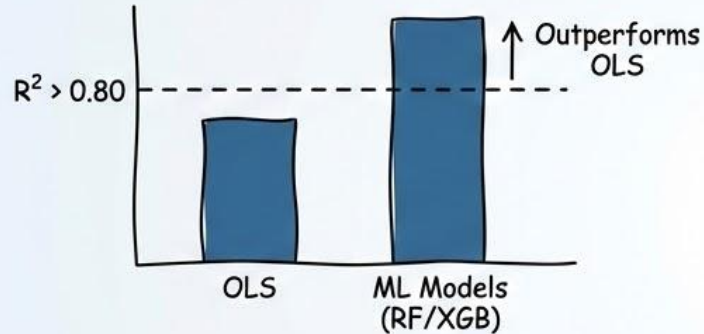
- **Baseline**: OLS Linear Regression (Interpretability). Count.
- **Challengers**: Random Forest & XGBoost (Non-linearity & Accuracy).
- **Tuning**: GridSearchCV for hyperparameter optimization.

# Expected Analytics Results

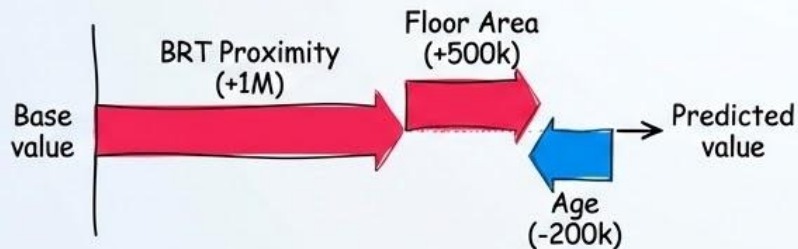
## Expected Analytics Results

### Visualizations & Examples

Model Accuracy Comparison (R-Squared)



SHAP Force Plot: Individual Valuation Explanation



e.g., "This condo is +1M due to BRT proximity"

### Key Expectations & Strategy



- **Accuracy Target:** Machine Learning models expected to achieve **R-Squared > 0.80**, outperforming OLS.



- **Key Drivers:** Expect 'Location' (Distance to IT Park) and 'Floor Area' to be top predictors.



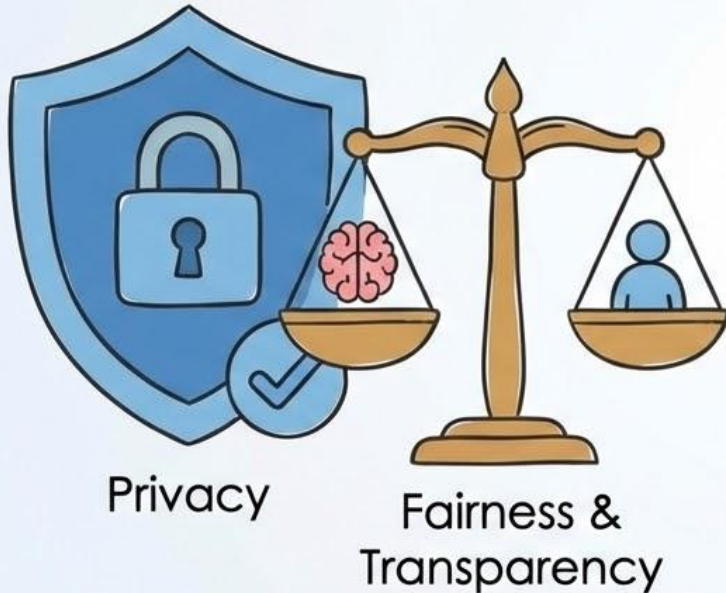
- **Explainability:** SHAP Force Plots will explain *individual* property valuations, providing transparent insights into model predictions.



# Ethical Considerations

## Ethical Considerations

### Ethical Framework & Compliance



### Key Principles & Actions



- **Data Privacy:** Strict adherence to DPA 2012. No private owner names or contact numbers will be scraped/stored.



- **Transparency:** The model is a *decision support tool*, not a replacement for professional appraisal.



- **Bias Mitigation:** 'Human-in-the-loop' validation with licensed brokers to flag realistic vs. speculative pricing.